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DATABASE SYSTEMS (NO:-3 HAPPINESSS DATA SET)

ASSESSMENT – 4

EXECUTION OF QUERIES:-

QUERY 1. SHOW DATABASES:-

COMMAND :-SHOW DATABASES;

OUTPUT:-

```
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| abhi     |
| frauddetectiondb |
| frauddetectionmanagementdb |
| frauddetectionmanagementdbms |
| frauddetectionmanagementdbmss |
| frauddetectionmanagementdbmsss |
| frauddetectionmanagementdbs |
| frauddetectionmanagementsystemdb |
| frauddetectionms |
| happiness_db |
| information_schema |
| mysql |
| performance_schema |
| productorderdb |
| sys |
+-----+
15 rows in set (0.02 sec)
```

QUERY 2 — FIND CSV FILE

OUTPUT:-

```
mysql> find ~/Downloads ~/Desktop ~/Documents -iname "2015.csv"
-> |
```

QUERY 3 — CREATE DATABASE

OUTPUT:-

```
mysql> CREATE DATABASE happiness_db;
ERROR 1007 (HY000): Can't create database 'happiness_db'; database exists
mysql> USE happiness_db;
Database changed
mysql> SELECT DATABASE();
+-----+
| DATABASE() |
+-----+
| happiness_db |
+-----+
1 row in set (0.00 sec)
```

QUERY 4 — CREATE TABLE

OUTPUT:-

```
mysql> CREATE TABLE happiness_2015 (  
-> Country VARCHAR(100),  
-> Region VARCHAR(100),  
-> Happiness_Rank INT,  
-> Happiness_Score DECIMAL(5,3),  
-> Standard_Error DECIMAL(10,5),  
-> Economy_GDP_Per_Capita DECIMAL(10,5),  
-> Family DECIMAL(10,5),  
-> Health_Life_Expectancy DECIMAL(10,5),  
-> Freedom DECIMAL(10,5),  
-> Trust_Government_Corruption DECIMAL(10,5),  
-> Generosity DECIMAL(10,5),  
-> Dystopia_Residual DECIMAL(10,5)  
-> );  
Query OK, 0 rows affected (0.08 sec)
```

QUERY 5 — CHECK LOCAL INFILE

OUTPUT:-

```
mysql> SHOW VARIABLES LIKE 'local_infile';  
+-----+-----+  
| Variable_name | Value |  
+-----+-----+  
| local_infile  | OFF   |  
+-----+-----+  
1 row in set (0.01 sec)  
  
mysql> SET GLOBAL local_infile = 1;  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> exit  
Bye
```

```
PS C:\Users\Asus> mysql --local-infile=1 -u root -p  
Enter password: *****  
Welcome to the MySQL monitor.  Commands end with ; or \g.  
Your MySQL connection id is 15  
Server version: 8.0.46 MySQL Community Server - GPL  
  
Copyright (c) 2000, 2026, Oracle and/or its affiliates.  
  
Oracle is a registered trademark of Oracle Corporation and/or its  
affiliates. Other names may be trademarks of their respective  
owners.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
mysql> SHOW VARIABLES LIKE 'local_infile';  
+-----+-----+  
| Variable_name | Value |  
+-----+-----+  
| local_infile  | ON    |  
+-----+-----+  
1 row in set (0.01 sec)
```

6. Enable Local Infile Settings

```
mysql> SET GLOBAL local_infile = 1;
Query OK, 0 rows affected (0.00 sec)
```

7. Import CSV Data

```
mysql> LOAD DATA LOCAL INFILE 'C:/Users/Asus/Downloads/2015.csv'
-> INTO TABLE happiness_2015
-> FIELDS TERMINATED BY ','
-> ENCLOSED BY '"'
-> LINES TERMINATED BY '\n'
-> IGNORE 1 ROWS;
Query OK, 158 rows affected (0.01 sec)
Records: 158 Deleted: 0 Skipped: 0 Warnings: 0
```

8. Total Record Count

```
mysql> SELECT COUNT(*) AS total_countries FROM happiness_2015;
+-----+
| total_countries |
+-----+
|             158 |
+-----+
1 row in set (0.00 sec)
```

9. . View Initial Data Samples

SELECT Country, Region, Happiness_Rank, Happiness_Score FROM
happiness_2015 LIMIT 10;

```
mysql> SELECT Country, Region, Happiness_Rank, Happiness_Score FROM happiness_2015 LIMIT 10;
+-----+-----+-----+-----+
| Country | Region | Happiness_Rank | Happiness_Score |
+-----+-----+-----+-----+
| Switzerland | Western Europe | 1 | 7.587 |
| Iceland | Western Europe | 2 | 7.561 |
| Denmark | Western Europe | 3 | 7.527 |
| Norway | Western Europe | 4 | 7.522 |
| Canada | North America | 5 | 7.427 |
| Finland | Western Europe | 6 | 7.406 |
| Netherlands | Western Europe | 7 | 7.378 |
| Sweden | Western Europe | 8 | 7.364 |
| New Zealand | Australia and New Zealand | 9 | 7.286 |
| Australia | Australia and New Zealand | 10 | 7.284 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

10. Regional Aggregations (GROUP BY & ORDER BY)

```
mysql> SELECT
->     Region,
->     ROUND(AVG(Happiness_Score), 2) AS avg_happiness_score,
->     ROUND(SUM(Economy_GDP_per_Capita), 2) AS total_gdp_contrib
-> FROM happiness_2015
-> GROUP BY Region
-> ORDER BY avg_happiness_score DESC;
```

Region	avg_happiness_score	total_gdp_contrib
Australia and New Zealand	7.29	2.58
North America	7.27	2.72
Western Europe	6.69	27.27
Latin America and Caribbean	6.14	19.29
Eastern Asia	5.63	6.91
Middle East and Northern Africa	5.41	21.34
Central and Eastern Europe	5.33	27.33
Southeastern Asia	5.32	7.10
Southern Asia	4.58	3.92
Sub-Saharan Africa	4.20	15.22

10 rows in set (0.00 sec)

DDL MODIFICATIONS

12&13 Inspect Table Structure

```
mysql> DESCRIBE happiness_2015;
```

Field	Type	Null	Key	Default	Extra
Country	varchar(100)	YES		NULL	
Region	varchar(100)	YES		NULL	
Happiness_Rank	int	YES		NULL	
Happiness_Score	decimal(5,3)	YES		NULL	
Standard_Error	decimal(10,5)	YES		NULL	
Economy_GDP_Per_Capita	decimal(10,5)	YES		NULL	
Family	decimal(10,5)	YES		NULL	
Health_Life_Expectancy	decimal(10,5)	YES		NULL	
Freedom	decimal(10,5)	YES		NULL	
Trust_Government_Corruption	decimal(10,5)	YES		NULL	
Generosity	decimal(10,5)	YES		NULL	
Dystopia_Residual	decimal(10,5)	YES		NULL	
Discount_Factor	decimal(5,2)	YES		NULL	

13 rows in set (0.01 sec)

```
mysql> SHOW COLUMNS FROM happiness_2015;
```

Field	Type	Null	Key	Default	Extra
Country	varchar(100)	YES		NULL	
Region	varchar(100)	YES		NULL	
Happiness_Rank	int	YES		NULL	
Happiness_Score	decimal(5,3)	YES		NULL	
Standard_Error	decimal(10,5)	YES		NULL	
Economy_GDP_Per_Capita	decimal(10,5)	YES		NULL	
Family	decimal(10,5)	YES		NULL	
Health_Life_Expectancy	decimal(10,5)	YES		NULL	
Freedom	decimal(10,5)	YES		NULL	
Trust_Government_Corruption	decimal(10,5)	YES		NULL	
Generosity	decimal(10,5)	YES		NULL	
Dystopia_Residual	decimal(10,5)	YES		NULL	
Discount_Factor	decimal(5,2)	YES		NULL	

13 rows in set (0.00 sec)

11. Add Discount Column

```
mysql> ALTER TABLE happiness_2015 ADD COLUMN Discount_Factor DECIMAL(5,2);
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

14. Sample Column Data

```
mysql> SELECT Country, Happiness_Score, Discount_Factor FROM happiness_2015 LIMIT 5;
```

Country	Happiness_Score	Discount_Factor
Switzerland	7.587	NULL
Iceland	7.561	NULL
Denmark	7.527	NULL
Norway	7.522	NULL
Canada	7.427	NULL

```
5 rows in set (0.00 sec)
```

15. Modify Column Data Type

```
mysql> ALTER TABLE happiness_2015 MODIFY COLUMN Happiness_Rank SMALLINT;
Query OK, 158 rows affected (0.06 sec)
Records: 158 Duplicates: 0 Warnings: 0
```

16&17. Execute Bulk Data Updates (UPDATE)

```
mysql> SELECT
-> Region,
-> ROUND(AVG(Happiness_Score), 2) AS avg_happiness_score,
-> ROUND(SUM(Economy_GDP_per_Capita), 2) AS total_gdp_contrib
-> FROM happiness_2015
-> GROUP BY Region
-> ORDER BY avg_happiness_score DESC;
```

Region	avg_happiness_score	total_gdp_contrib
Australia and New Zealand	7.29	2.58
North America	7.27	2.72
Western Europe	6.69	27.27
Latin America and Caribbean	6.14	19.29
Eastern Asia	5.63	6.91
Middle East and Northern Africa	5.41	21.34
Central and Eastern Europe	5.33	27.33
Southeastern Asia	5.32	7.10
Southern Asia	4.58	3.92
Sub-Saharan Africa	4.20	15.22

```
10 rows in set (0.00 sec)
```

```
mysql> UPDATE happiness_2015
-> SET Discount_Factor = CASE
-> WHEN Region = 'Western Europe' THEN ROUND(RAND() * 15, 2)
-> WHEN Region = 'North America' THEN ROUND(RAND() * 10, 2)
-> ELSE ROUND(RAND() * 5, 2)
-> END;
Query OK, 158 rows affected (0.01 sec)
Rows matched: 158 Changed: 158 Warnings: 0
```

18. Rename Column

```
mysql> ALTER TABLE happiness_2015 RENAME COLUMN Region TO World_Region;
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

19. Add Table Check Constraint

```
mysql> ALTER TABLE happiness_2015 ADD CONSTRAINT chk_score CHECK (Happiness_Score >= 0);
Query OK, 158 rows affected (0.05 sec)
Records: 158 Duplicates: 0 Warnings: 0
```

20. Rename Table

```
mysql> RENAME TABLE happiness_2015 TO happiness_transactions;  
Query OK, 0 rows affected (0.02 sec)
```

21. Create Lookup Index

```
mysql> CREATE INDEX idx_world_region ON happiness_transactions (World_Region);  
Query OK, 0 rows affected (0.04 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

DCL COMMANDS:-

22. Create User

```
mysql> CREATE USER 'happiness_analyst'@'localhost' IDENTIFIED BY '123456';  
Query OK, 0 rows affected (0.02 sec)
```

23&27 Grant Permissions

```
mysql> GRANT SELECT ON happiness_db.happiness_transactions TO 'happiness_analyst'@'localhost';  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> GRANT SELECT, INSERT ON happiness_db.happiness_transactions TO 'happiness_analyst'@'localhost';  
Query OK, 0 rows affected (0.00 sec)  
  
mysql> GRANT SELECT, INSERT, UPDATE ON happiness_db.happiness_transactions TO 'happiness_analyst'@'localhost';  
Query OK, 0 rows affected (0.00 sec)  
  
mysql> GRANT ALL PRIVILEGES ON happiness_db.happiness_transactions TO 'happiness_analyst'@'localhost';  
Query OK, 0 rows affected (0.00 sec)  
  
mysql> GRANT ALL PRIVILEGES ON happiness_db.* TO 'happiness_analyst'@'localhost';
```

28. Apply Privilege Updates

```
mysql> FLUSH PRIVILEGES;  
Query OK, 0 rows affected (0.01 sec)
```

29. Check Active User Grants

```
mysql> SHOW GRANTS FOR 'happiness_analyst'@'localhost';  
+-----+  
| Grants for happiness_analyst@localhost |  
+-----+  
| GRANT USAGE ON *.* TO 'happiness_analyst'@'localhost' |  
| GRANT ALL PRIVILEGES ON 'happiness_db'.* TO 'happiness_analyst'@'localhost' |  
| GRANT ALL PRIVILEGES ON 'happiness_db'.'happiness_transactions' TO 'happiness_analyst'@'localhost' |  
+-----+  
3 rows in set (0.00 sec)
```

30&32. Revoke Grants & Drop User

```
mysql> REVOKE UPDATE, DELETE ON happiness_db.happiness_transactions FROM 'happiness_analyst'@'localhost';  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> REVOKE ALL PRIVILEGES ON happiness_db.* FROM 'happiness_analyst'@'localhost';  
Query OK, 0 rows affected (0.00 sec)  
  
mysql> DROP USER 'happiness_analyst'@'localhost';  
Query OK, 0 rows affected (0.01 sec)
```